

Shield-AI

47
TESTS PASSING

It doesn't predict the market.
It defends the downside.

SAME DRAWDOWN, TWO OUTCOMES



99% Portfolio VaR

GARCH(1,1)

Kupiec · Christoffersen

Alpaca MCP execution

THE PROBLEM

Confidence is not a risk framework.

Hand a language model a brokerage key and tell it to pick winners, and you have not built a quant system. You have built a leveraged guess.

1

No covariance matrix

Exposures across names cannot be added up

2

No tail estimate

Crash risk is invisible to the model

3

No loss budget

Nothing defines how much is too much

“

**The model is confident,
not calibrated.**

Shield-AI inverts the design: deterministic mathematics defines the boundary, and the language model only ever operates inside it.

The scoring rule decides the architecture.

Performance is judged on terminal account equity. There is no benchmark column.

5

trading days in the scoring window

30%

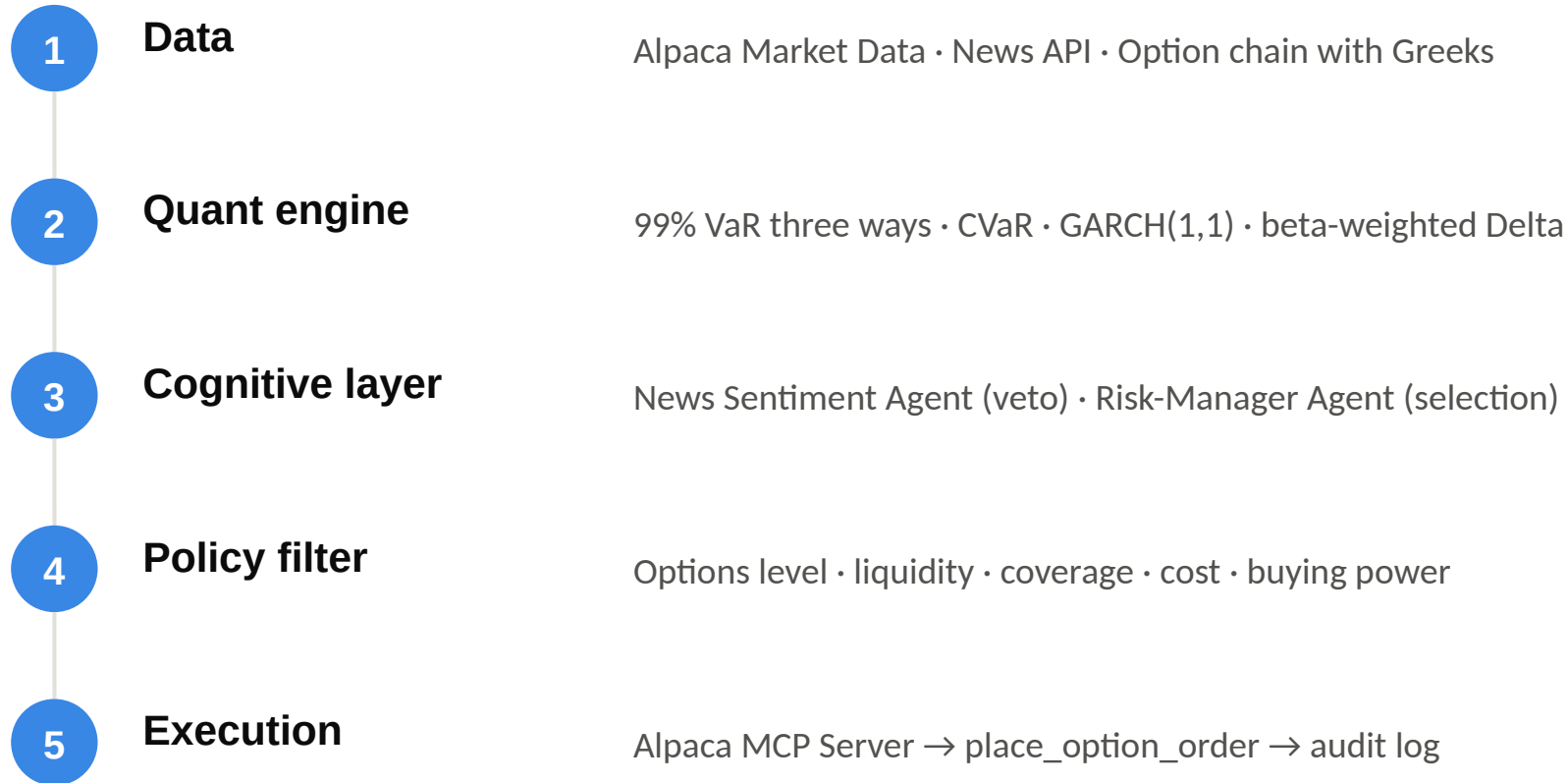
drawdown that cannot be recovered in
that time

0

benchmarks — only the final number
counts

Maximising final equity is therefore a risk-management problem, not a prediction problem.

Five layers, one boundary.



Execution runs over MCP — an open protocol, not a hard-coded SDK binding. The same agent could be pointed at any MCP-compliant venue.

THE STRATEGY

One model, two jobs.

GARCH forecasts realised volatility. Options imply their own. The gap between them is the variance risk premium — and it is both the risk metric and the trading signal.

IV $\gg$ forecast

Options are rich

**Sell premium
cash-secured put**

IV $\ll$ forecast

Options are cheap

**Buy protection
long put**

VaR breached

Risk event

**Hedge — regardless
of how rich the premium**

Order of precedence is the thesis: a VaR breach outranks any income opportunity, however rich.

Sizing is solved for, never chosen — a position may consume at most half the remaining headroom to the limit.

GARCH cannot read that the FDA decision is on Thursday.

Selling volatility into a known catalyst is a classic way to destroy an account. Implied volatility is elevated because the market knows something is coming — so the "edge" is not an edge. It is compensation for a risk the volatility model structurally cannot perceive.

**The agent holds a VETO,
never a vote.**

It proposes nothing and sizes nothing. It answers one question: is there an unresolved catalyst inside the life of this option?

WHAT THE AGENT CAUGHT

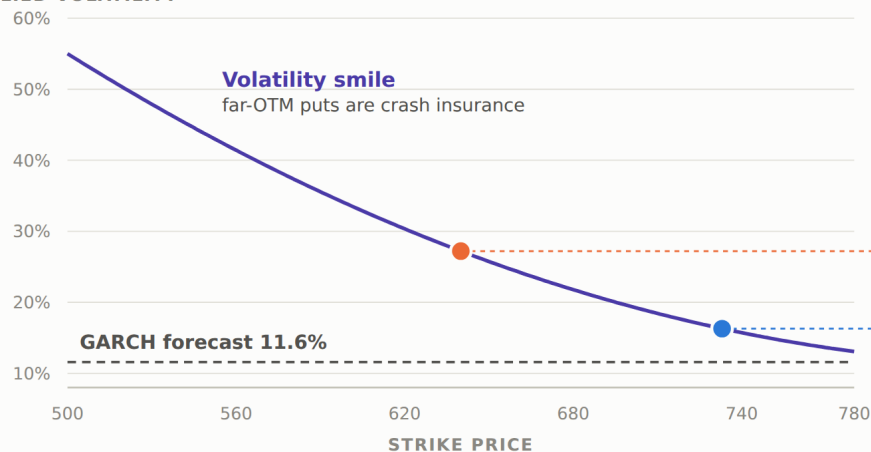
It found a 15.6-point edge. It was my bug.

SHIELD-AI • VARIANCE RISK PREMIUM

My agent found a 15.6 vol-point edge on SPY. It was my bug.

Implied volatility across the SPY put chain, 7 days to expiry

IMPLIED VOLATILITY



Volatility smile

far-OTM puts are crash insurance

✗ Median of whole chain

27.1% → phantom edge +15.6 pts

✓ Delta-matched contract

16.3% → real edge +4.7 pts

The lesson: a summary statistic across the chain measures skew, not premium. Compare a forecast only to the contract you would actually trade.

I compared the GARCH forecast to the MEDIAN implied volatility across the whole option chain. But far out-of-the-money puts trade at much higher IV — they are crash insurance. The median sat on that wing.

I was measuring skew and calling it alpha.

Fixed to read IV from a single delta-matched contract. Now guarded by a regression test.

EVIDENCE

A log of refusals proves judgement.

47

verification tests passing

3

independent VaR estimators

2

formal backtests: Kupiec + Christoffersen

First live cycle, 1 September 2026

● NVDA

GARCH 41.9% vs IV 29.8% → protection 12 points cheap. Bought 4 protective puts, \$396 of a \$1,500 budget. Order 6c693bae...

● 4 refusals

Edge inside the no-trade band, or open interest too thin to trust the quote. Every refusal logged with its reason.

A record of only executed trades proves an agent traded. A record of refusals proves it had judgement.

**Risk management — not alpha —
is the killer app for agentic finance.**

Shield-AI · github.com/amina178/shield-ai

Lablab.ai × Alpaca AI Trading Agents Hackathon · Agent Builder track